SYED HUZAFA ALI

AI / ML ENGINEER — RETRIEVAL-AUGMENTED GENERATION (RAG) SYSTEMS

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Summary

AI Engineering specializing in production-grade Retrieval-Augmented Generation (RAG) and LLM-powered systems. Experienced in building end-to-end pipelines using Python, FastAPI, LangChain, and vector databases (Pinecone, FAISS, Chroma), with hands-on work in hybrid search, query transformation, and re-ranking to improve retrieval accuracy. Comfortable taking AI applications from prototype to deployable backend using Docker and AWS. Seeking an AI/ML Engineer role to apply RAG, LLM integration, and full-stack data engineering skills to real-world products.

Technical Skills

Languages: Python, SQL
AI / LLM Engineering: RAG Pipelines, LangChain, Prompt Engineering, OpenAI GPT-4o API, Claude API, Hybrid Search, Query Transformation, Cross-Encoder Re-ranking, Contextual Compression
Frameworks & Tools: FastAPI, Streamlit, Docker, Git, GitHub
Libraries: NumPy, Pandas, Matplotlib, Pydantic
Databases: Pinecone, FAISS, Chroma (Vector DBs), MySQL
Cloud: AWS (EC2, S3 — basic)

Projects

Multimodal RAG System with Hybrid Retrieval *Python, LangChain, Pinecone, FastAPI, OpenAI GPT-4o*

- Built a multimodal RAG pipeline handling unstructured text, tables, and embedded images in PDFs, using GPT-4o for cross-modal synthesis.
- Designed hybrid retrieval combining Pinecone dense embeddings with BM25 sparse search via Reciprocal Rank Fusion (RRF); built a 50-query evaluation set and improved Precision from 68% (vector-only) to 81% (hybrid).
- Added multi-query expansion to improve retrieval across paraphrased user prompts, increasing Recall@5 by 13% on the same evaluation set.
- Applied a cross-encoder re-ranker on top-k retrieved documents, reducing irrelevant context by 26% and lowering average LLM token usage by 18% per query.
- Used contextual compression to retain only relevant spans, reducing average prompt size from 2,100 to 1,350 tokens (36% reduction) with no measurable loss in answer quality.
- Deployed the system as an async FastAPI backend with a Streamlit frontend, achieving p50 latency of 1.6 s and p95 latency of 3.4 s under 20 concurrent users.

NocodeNexus — No-Code Data Preprocessing Platform (Final Year Project) *Python, Pandas, NumPy, React, Node.js, FastAPI*

- Built a full-stack, automated ETL platform enabling non-technical users to clean, normalize, and validate structured datasets without writing code.
- Improved data processing speed by 30% by replacing iterative loops with vectorized NumPy/Pandas operations, enabling efficient handling of high-concurrency workloads.
- Designed and documented RESTful APIs with FastAPI to connect the React frontend with the backend processing engine, supporting modular ETL operations (cleaning, normalization, validation).

Education

University of Engineering and Technology (UET), Lahore 2022 – 2026
Bachelor of Science in Computer Science — CGPA: 3.08/4.0
Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming

Government College of Science, Lahore 2019 – 2021
Intermediate in Computer Science (ICS) — Score: 83%

Certifications

- Python for Data Science, AI & Development — IBM
- Machine Learning with Python — IBM
- Rag & Agentic Ai Professional Certificate — IBM
- Python Programming — IBM

LANGUAGES English, Urdu